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Natural Language Processing Lab 3

Dr. Chetana Gavankar

Associate Professor
Birla Institute of Technology & Science, Pilani

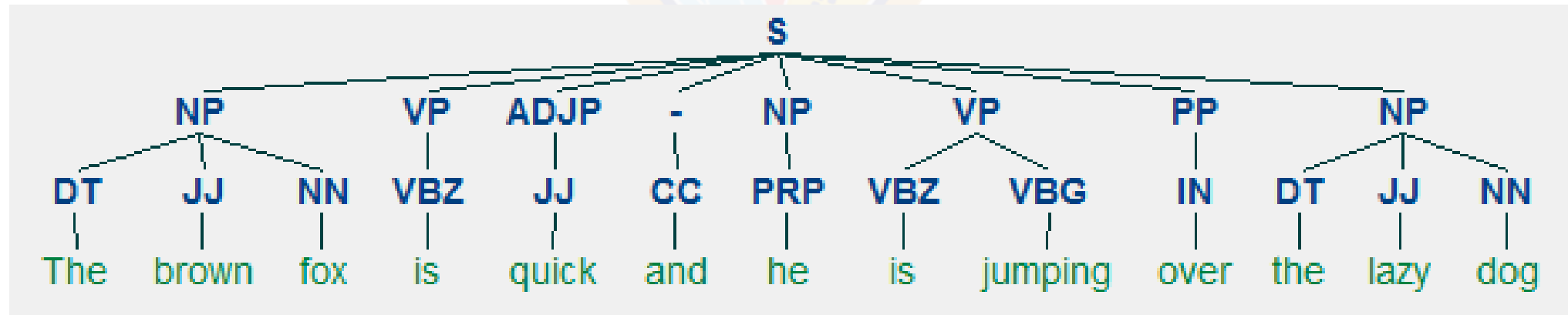
In this Lab Demo we will cover:

- Parsing
- Context Free Grammar
- Chart Parsing
- Probabilistic Context Free Grammar
- Viterbi Parser



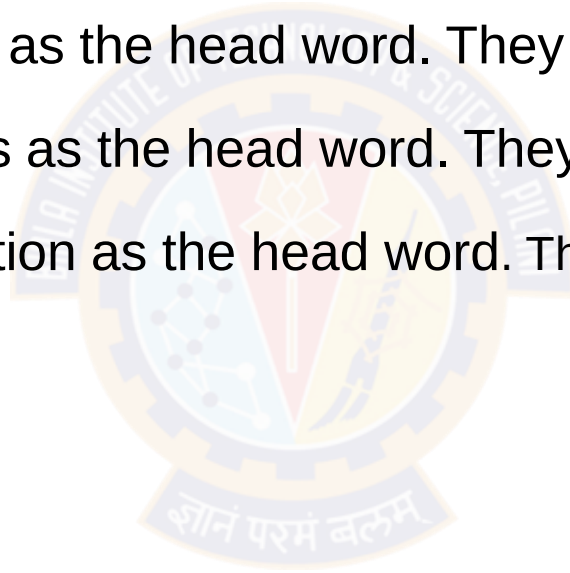
Parsing

- Analysis of words in the sentence for grammar and arranging words in a manner that shows the relationship among the words.
- Analyzing the structure of a sentence to break it down into its smallest constituents (which are tokens such as words) and group them together into higher-level phrases.
- This includes POS tags as well as phrases from a sentence.



Categories of Phrases

- **Noun phrase (NP):** Noun acts as the head word. They start with an article or noun.
- **Verb phrase (VP):** Verb acts as the head word. They start with an verb
- **Adjective phrase (ADJP):** Adjective as the head word. They start with an adjective
- **Adverb phrase (ADVP):** Adverb acts as the head word. They usually start with an adjective
- **Prepositional phrase (PP):** Preposition as the head word. They start with an preposition.



Context Free Grammar

$$(1) \ S \rightarrow NP \ VP$$

$$(2) \ NP \rightarrow ART \ ADJ \ N$$

$$(3) \ NP \rightarrow ART \ N$$

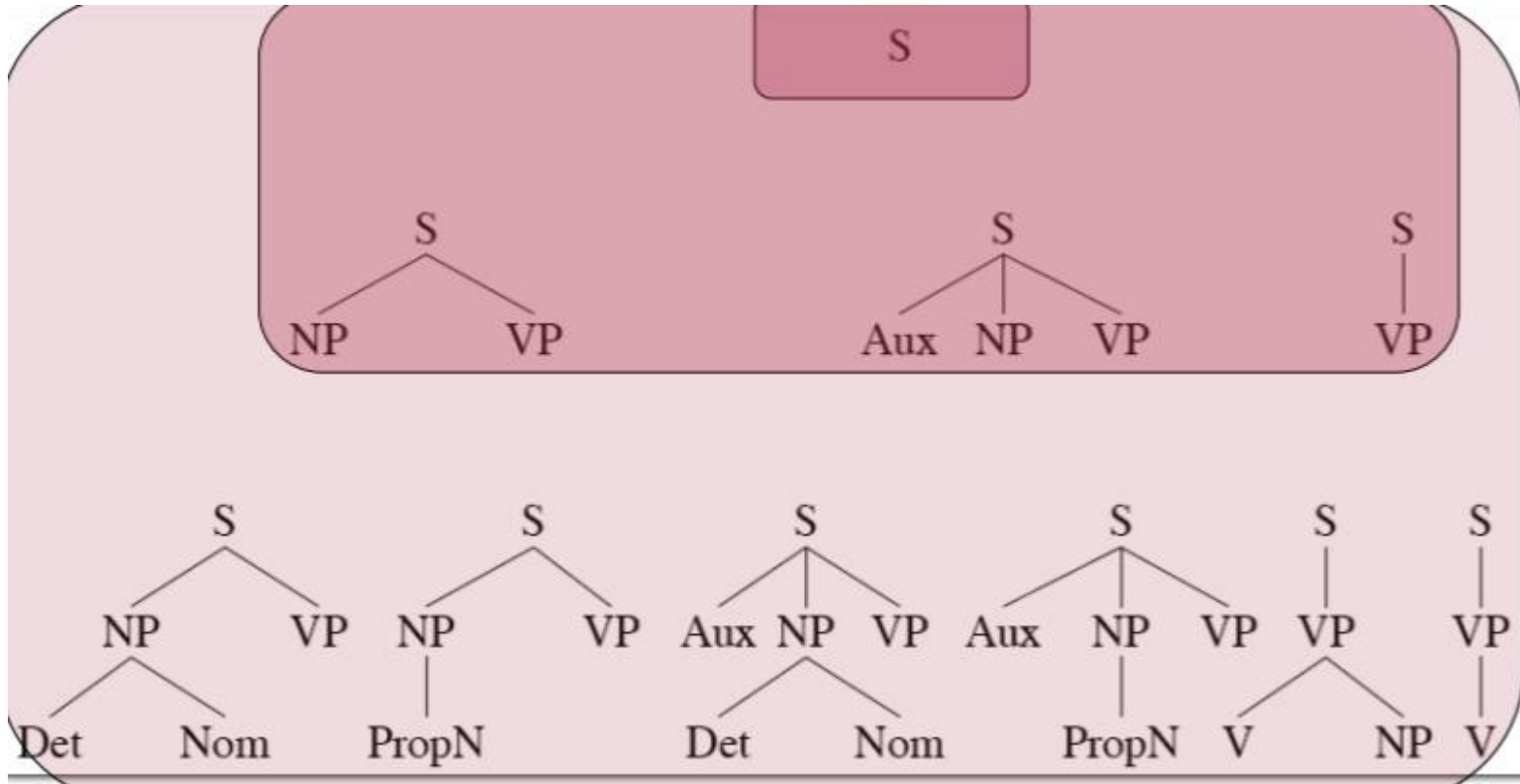
$$(4) \ NP \rightarrow ADJ \ N$$

$$(5) \ VP \rightarrow AUX \ VP$$

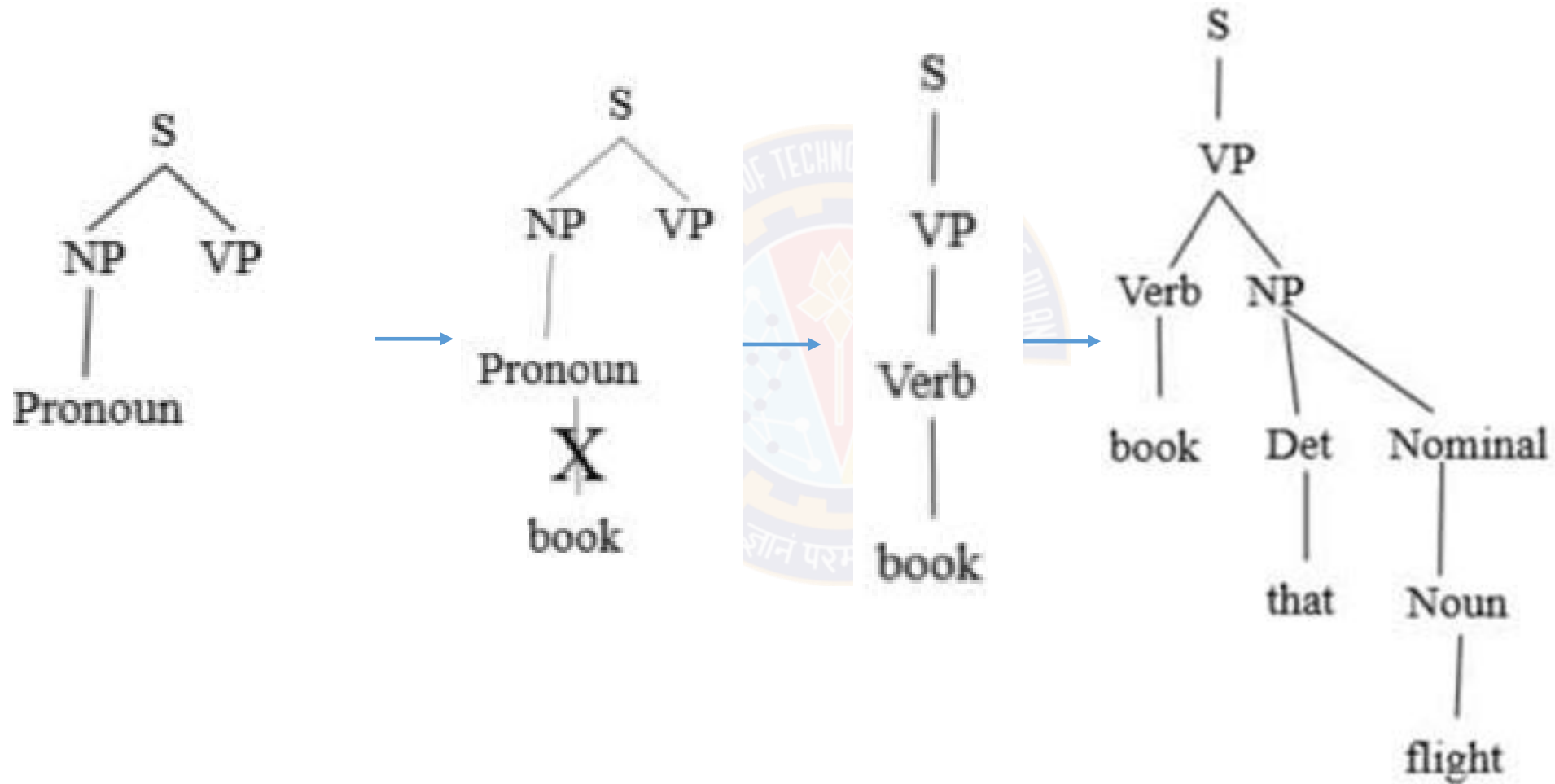
$$(6) \ VP \rightarrow V \ NP$$

Top down Parsing

- **Parse tree is generated in the top to bottom fashion**
- **from root to leaves**
- **Search space from the start symbol sentence S on LHS**



Top down Parsing



Bottom up Parsing

- Bottom-up parsing method is just inverse of top-down
- Input string is taken in the first place and then the string is reduced using the grammar.
- In this process the start symbol is acquired at the end of the process

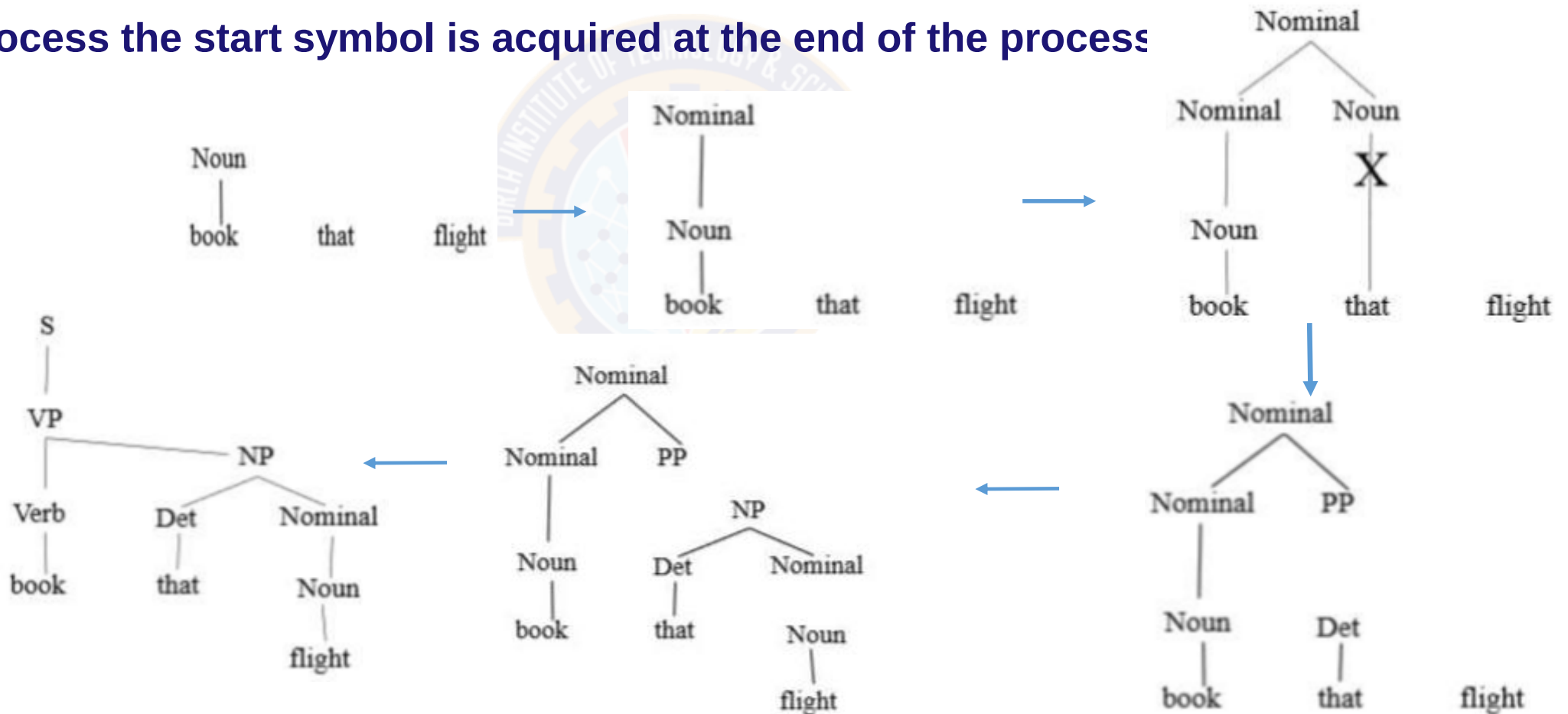


Chart Parsing

- When a chart parser begins parsing a text, it creates a new (empty) chart, spanning the text.
- It then incrementally adds new edges to the chart.
- A set of "chart rules" specifies the conditions under which new edges should be added to the chart.
- Once the chart reaches a stage where none of the chart rules adds any new edges, parsing is complete.
- **Advantages of Chart Parsing**
 - No repeated computation of same sub problem
 - Deals well with left-recursive grammars
 - Deals well with ambiguity
 - No backtracking necessary

Chart Parsing Example

The input: “ x = The large can can hold the water”

POS of Input Words:

- the: *ART*
- large: *ADJ*
- can: *N*, *AUX*, *V*
- hold: *N*, *V*
- water: *N*, *V*

The input: “ x = **The** large can can hold the water”

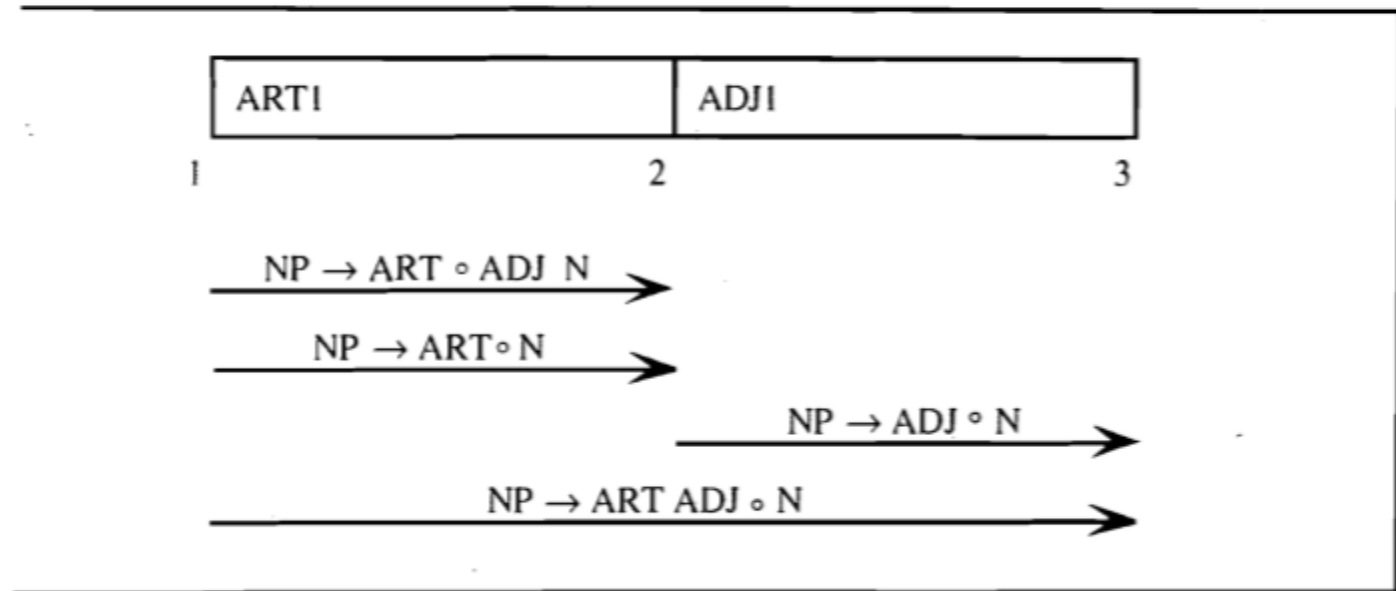
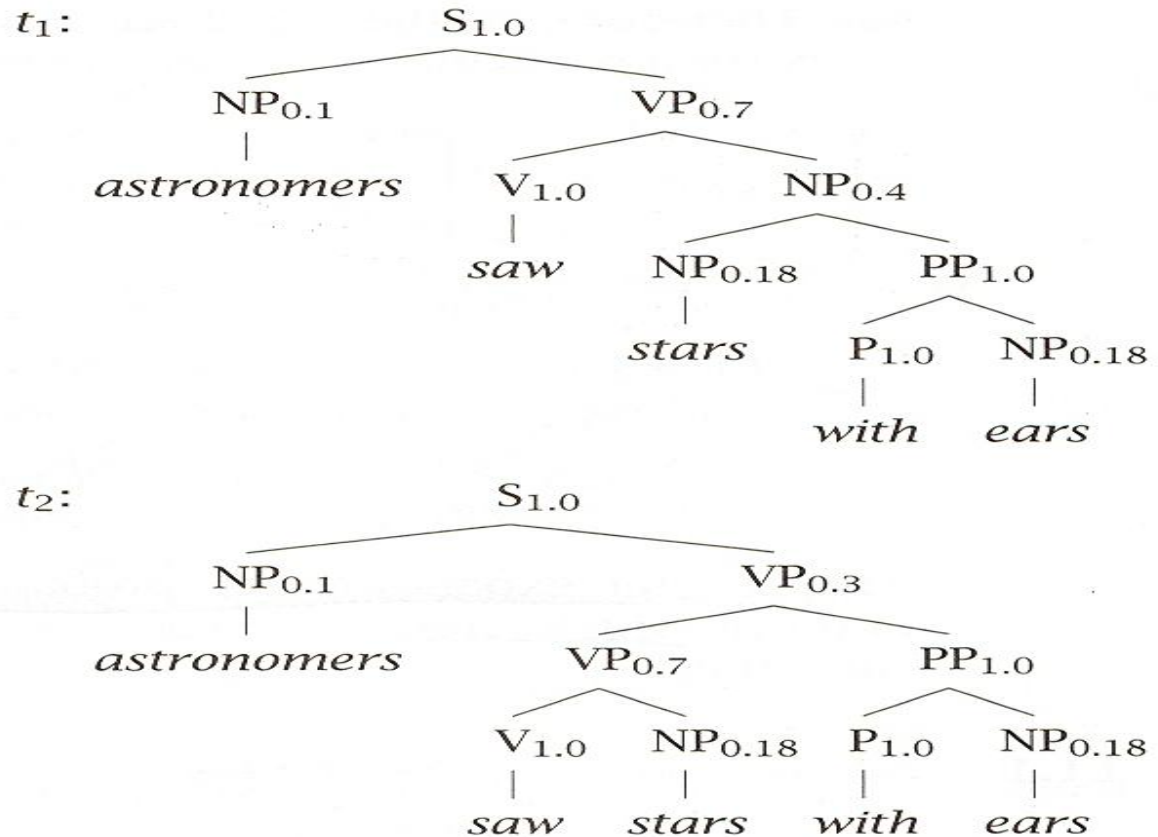


Figure 3.9 The chart after seeing an ADJ in position 2

Probabilistic Context Free Grammar



- Each CFG rule has associated probability
- How likely it is that this production will be used

$S \rightarrow NP VP$	1.0	$NP \rightarrow NP PP$	0.4
$PP \rightarrow P NP$	1.0	$NP \rightarrow \textit{astronomers}$	0.1
$VP \rightarrow V NP$	0.7	$NP \rightarrow \textit{ears}$	0.18
$VP \rightarrow VP PP$	0.3	$NP \rightarrow \textit{saw}$	0.04
$P \rightarrow \textit{with}$	1.0	$NP \rightarrow \textit{stars}$	0.18
$V \rightarrow \textit{saw}$	1.0	$NP \rightarrow \textit{telescopes}$	0.1

NLTK POS Tagging Lab Demo

- Write context free grammar rules for a given sentence
- Implement Recursive descent parser using NLTK in python
- Implement Chart parser using NLTK in python
- Implement Probabilistic context free grammar using inside chart parser for a sentence from NLTK corpus
- Implement Probabilistic context free grammar using Viterbi algorithm for a sentence from NLTK corpus
- Draw Parse tree
- Identify whether the sentence is ambiguous or unambiguous



Thank you